Variance-Corrected Multi-Asset Equity Simulation with Hybrid Hidden Markov Marginals

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CRediT authorship contribution statement

Abdulrahman Alswaidan: Conceptualization, Methodology, Software, Investigation, Validation, Writing - original draft, Writing - review and editing. **Jeffrey D. Varner:** Conceptualization, Methodology, Software, Investigation, Validation, Writing - original draft, Writing - review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Code and data availability

The experiment scripts, evaluation code (six-method comparison, left-tail Value-at-Risk coverage check, and bias correction), cached inputs and result summaries, and instructions for regenerating the fitted per-asset models are available in the paper repository. The underlying model-fitting library is released in the JumpHMM.jl repository and is pinned as a dependency of the experiment code.

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